PAPER GRANNY
arXiv Paper Interpretation Agent

Denoising Diffusion Probabilistic Models

Deep Interpretation Report

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Paper Information

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Key Contribution: High-quality image synthesis using a simplified diffusion model objective.

目录

1 Background: The Art of Adding and Removing Noise **2**

2 Core Problem and Innovation **2**

2.1 The Challenge 2

2.2 The Innovation 2

3 Methodology: How it Works **3**

3.1 The Forward Process 3

3.2 The Reverse Process (The "Magic") 3

3.3 Sampling (Generating New Images) 3

4 Experimental Results **4**

4.1 Progressive Generation 4

5 Summary and Conclusion **5**

6 References **6**

1 Background: The Art of Adding and Removing Noise

What is a Diffusion Model?

Imagine you have a clear photograph. If you sprinkle a little bit of salt-and-pepper noise on it, it gets slightly blurry. If you keep doing this for 1000 steps, eventually the photo becomes pure static—just random noise.

A **Diffusion Model** is a machine learning technique that learns to do the exact opposite: it starts with a screen full of random noise and tries to "clean" it step-by-step until a beautiful, clear image emerges.

Technically, this process involves two stages:

- **Forward Process (Diffusion)**: Gradually adding Gaussian noise to data until it becomes pure noise. This is a fixed process with no learning involved.
- **Reverse Process (Denoising)**: A learned neural network that tries to predict how to remove the noise added in the forward process.

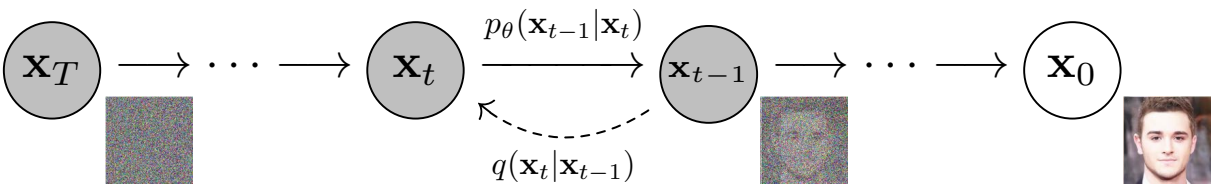

图 1: The directed graphical model of the diffusion process. $\mathbf{x}_0$ is the data, and $\mathbf{x}_T$ is pure noise.

2 Core Problem and Innovation

2.1 The Challenge

Before this paper, diffusion models existed but weren't known for high-quality image generation. They were often seen as mathematically elegant but practically inferior to Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs).

2.2 The Innovation

The authors introduced several key changes that made diffusion models "state-of-the-art":

Key Innovations

- **Noise Prediction**: Instead of trying to predict the "clean" image directly at each step, the network is trained to predict the *noise* that was added. This makes the learning task much easier.
- **Simplified Objective**: They found that a simple Mean Squared Error (MSE) loss between the true noise and the predicted noise works better than the complex mathematical "variational bound" used previously.
- **U-Net Architecture**: Using a powerful U-Net with self-attention and sinusoidal time embeddings allowed the model to handle complex image structures.

3 Methodology: How it Works

3.1 The Forward Process

The forward process $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ adds a small amount of Gaussian noise at each step t:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

A beautiful property of this process is that we can jump from the original image $\mathbf{x}_0$ to any noisy step $\mathbf{x}_t$ directly:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

where $\bar{\alpha}_t$ is a product of noise-related constants.

3.2 The Reverse Process (The "Magic")

The neural network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ takes the noisy image $\mathbf{x}_t$ and the current time step t as input. Its goal is to predict the noise $\boldsymbol{\epsilon}$ that was used to create $\mathbf{x}_t$.

Training Algorithm

1. Pick a clean image $\mathbf{x}_0$ from the dataset. 2. Pick a random time step t between 1 and 1000. 3. Generate random noise $\boldsymbol{\epsilon}$. 4. Create a noisy image $\mathbf{x}_t$ using $\mathbf{x}_0, t,$ and $\boldsymbol{\epsilon}$. 5. Ask the network to predict $\boldsymbol{\epsilon}$ from $\mathbf{x}_t$ and t. 6. Update the network to minimize the difference: $\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2$.

3.3 Sampling (Generating New Images)

To generate a new image, we start with pure noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and repeatedly apply the reverse step:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$$

where $\mathbf{z}$ is a bit of fresh noise to keep the process stochastic.

4 Experimental Results

图 2: High-quality samples generated on CelebA-HQ 256×256.

4.1 Progressive Generation

One interesting finding is that the model generates images "coarse-to-fine". In the early steps of the reverse process, the model decides on the general shapes and colors. In the final steps, it fills in the fine details like skin texture or hair strands.

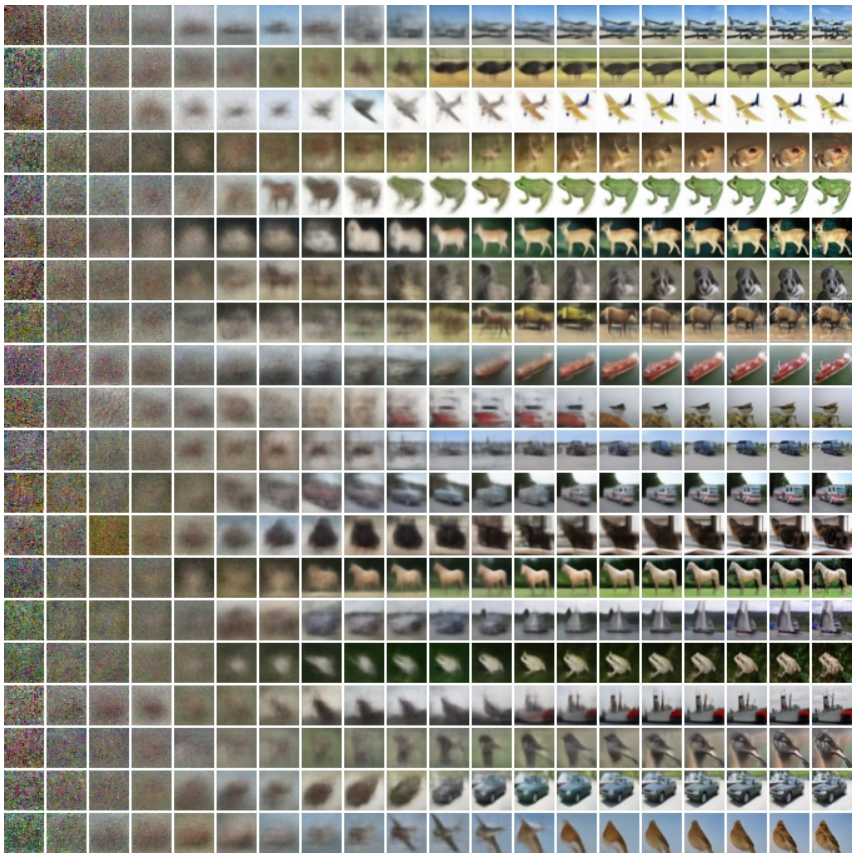

图 3: Progressive generation: from noise to a clear image (left to right).

5 Summary and Conclusion

Key Takeaways:

- **Simplicity Wins**: Predicting noise with a simple MSE loss is highly effective.
- **New SOTA**: Diffusion models are now a top-tier choice for generative modeling, rivaling GANs in quality.
- **Mathematical Bridge**: This work connects diffusion models to "score matching" and "Langevin dynamics," providing a solid theoretical foundation.

Limitations

The main drawback of DDPM is its **slow sampling speed**. Since it requires 1000 steps of neural network evaluations to generate a single image, it is much slower than GANs or VAEs which can generate images in a single pass.

6 References

- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. *Advances in Neural Information Processing Systems (NeurIPS)*.

- Original Code: `https://github.com/hojonathanho/diffusion`